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RESEARCH ARTICLE

The Drug-Like Molecule Pre-Training Strategy for Drug Discovery

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ABSTRACT Recent advances in artificial intelligence (AI) have led to the development of transformer-based models that have shown success in identifying potential drug molecules for therapeutic purposes. However, for a molecule to be considered a viable drug candidate, it must exhibit certain desirable properties such as low toxicity, high druggability, and synthesizability. To address this, we propose an approach that incorporates prior knowledge about these properties during the model training process. In this study, we utilized the PubChem database, which contains 100 million molecules, to filter drug-like molecules based on the quantity of drug-likeness (QED) score and the Pfizer rule. We then used this filtered dataset of drug-like molecules to train both molecular representation (ChemBERTa) and molecular generation models (MolGPT). To assess the performance of the molecular representation model, we fine-tuned the results on the MoleculeNet benchmark datasets. Meanwhile, we evaluated the performance of the molecular generation model based on the generated samples comprising 10,000 molecules. Despite the limited diversity of the pre-training dataset, the models for molecular representation were able to retain at least 90% of their original performance on benchmark datasets, with an additional improvement of 6% in predicting clinical toxicology. In the domain of molecular generation, the model pre-trained on drug-like molecules exhibited a high rate of desirable molecule properties in the unconditionally generated outputs. Additionally, the diversity of generated structures demonstrated notable performance compared to the conditional generation approach. Moreover, the drug-like molecule pre-training strategy is not limited to a specific model or training method, making it a flexible approach that can be easily modified based on the research interests and criteria of interest.

INDEX TERMS AI-based drug discovery, pre-training, quantity of drug-likeness, Pfizer rule.

I. INTRODUCTION

The transformer architecture has become a popular approach for processing the molecular sequence due to the outstanding performances in molecular sequence representation, and generation. With the self-attention mechanism and pre-training approach, the molecular sequence could be encoded as a meaningful vector in a feature space [1], [2]. Recent studies have demonstrated that the transformer-based model has shown outstanding performance in predicting drug-target interaction [3], [4], molecular properties [1], [5], [6], [7], and drug-drug interactions [8]. Similarly, in molecular sequence

generation, the transformer-based model has shown considerable performance in conditional and unconditional molecular sequence generation [2] compared to conventional generative models, such as variational auto-encoder (VAE) [9], [10], adversarial auto-encoder (AAE) [11], [12], [13], [14], and generative adversarial network (GAN) [15], [16], [17].

The training process for a transformer-based model typically comprises two primary stages: pre-training and fine-tuning. Fine-tuning specifically pertains to the learning process focused on the task that serves as the primary objective, such as drug toxicity prediction. In contrast, pre-training serves as an initial training step aimed at improving the model's general performance prior to specific target tasks. When a model is directly trained on a target task with limited

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labeled data, its ability to comprehend the meaning of a given sequence is naturally constrained. In the literature, various papers have adopted pre-training as an intermediate step to facilitate the acquisition of the semantic and functional characteristics embedded in the sequence [18], [19], [20]. Pre-training involves training the model on a large amount of unlabeled data. The underlying assumption is that the pre-training phase included a wide range of diverse data, which can subsequently benefit the fine-tuning process. The idea is that this approach enables the comprehensive acquisition of the various features present in the sequence.

In order to train a model capable of processing molecule sequences, the simplified molecular input line entry system (SMILES) strings were utilized as inputs. SMILES, similar to natural language, possesses both functional and semantic meanings, enabling the transformer-based model to learn the underlying features inherent in the given SMILES sequences. In this pre-training stage, large quantities of molecules obtained from publicly accessible molecule databases, including ZINC, PubChem, and ChEMBL, are commonly employed [21], [22], [23]. Certain studies have proposed subsets of molecule databases derived from the aforementioned public databases, such as MOSES and GuacaMol, which are predominantly utilized in molecular sequence generation tasks [24], [25].

Nevertheless, Krishna et al. conducted a study that revealed an important observation: the inclusion of a data-driven pre-training phase encompassing extensive and diverse features does not consistently yield favorable outcomes during the subsequent fine-tuning process [26]. Their findings indicate that enhancing target performance in the fine-tuning phase does not necessarily rely on the incorporation of extensive and diverse data. Instead, the key lies in pre-training methods that consider the distribution of the target data, which can provide notable advantages. In accordance with Krishna's argument, we have determined that it is imperative to reevaluate the pre-training process for the molecular sequence model. Rather than solely relying on pre-training with extensive and diverse datasets, we aim to optimize the advantages of pre-training specifically tailored to the target task. Our approach involves constructing and training subsets of data that align with desirable molecular properties. Through this strategy, we can implicitly teach knowledge regarding desirable molecular properties during the pre-training phase of the model.

This study aims to shed light on the impact of a pre-training dataset filtered by several desirable molecular filters. To accomplish this, we have chosen well-established and publicly available models that are specifically designed for molecular representation (ChemBERTa) and molecular generation (MolGPT). Both of these models are based on the transformer architecture. We selected both models as a baseline due to the publicly accessible code and weight parameters, and suitable performances for the molecular representation and generation.

II. METHODS

Figure 1 illustrates the conceptual diagram outlining our proposed approach. The drug-like molecule pre-training strategy involved training a model that specifically focuses on the desirable molecule spaces. The objective of drug-like molecule pre-training is to facilitate the implicit transfer of knowledge regarding desirable properties to the model while keeping the model's structures and training procedures unchanged.

A. DATASETS

In this study, we collected 100 million SMILES sequences from the PubChem database to calculate the quantity of drug-likeness (QED) score. After applying a filter to the dataset, we obtained 42 million drug-like molecular sequences with a QED score threshold of 0.7. The resulting filtered dataset was defined as the QED-filtered dataset, which included QED-filtered drug-like molecules (DM-Q). Within the DM-Q dataset, we applied the Pfizer rule, a molecule property-based rule for assessing toxicity [27], [28], to filter molecules and obtain the DM-QP dataset comprising 13 million sequences out of 42 million.

For molecular representation learning, we randomly sampled 10 million molecules from the DM-Q and DM-QP datasets. Since the performance of the original ChemBERTa model was related to the size of the pre-training dataset, we limited the pre-training dataset to the same size as the original ChemBERTa pre-training dataset [1].

The MOSES and GuacaMol datasets were selected as benchmark datasets for drug generation. These datasets are based on the ZINC and ChEMBL molecule databases and consist of 1.7 M and 1.3 M molecules, respectively. They are widely used in the field of drug discovery and serve as standardized benchmarks for evaluating the performance of molecular generation models. We trained a generative model based on these two datasets, and compared its performance to the generative model trained using the DM-Q and DM-QP datasets.

B. MOLECULAR REPRESENTATION

In this study, we employed ChemBERTa, which is a molecular variant of the RoBERTa model [1], [7]. RoBERTa is a transformer encoder-based sequence model commonly known as a BERT-like model, renowned for its proficiency in sequence representation [29], [30]. During the pre-training phase, we utilized the masked language modeling (MLM) technique [29]. MLM involves reconstructing the original sequence from a version in which certain tokens have been masked. The objective of the MLM task is to accurately restore the original sequence by inferring the correct tokens from the remaining ones. Through this restoration process, the model can effectively learn how to identify the appropriate tokens that were masked or corrupted. The ChemBERTa model has a maximum sequence length limitation of

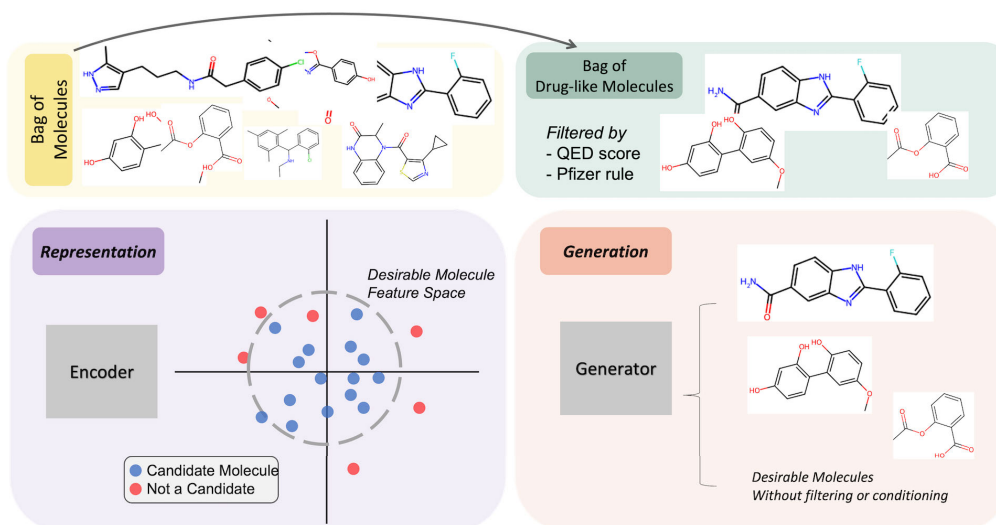


FIGURE 1. Conceptual diagram of the drug-like molecule pre-training for molecular representation and molecular generation.

512 tokens. It consists of 6 layers and encompasses approximately 44.5 million parameters.

The original ChemBERTa models were pre-trained on SMILES sequences of different sizes, including 5 M, 10 M, and 77 M [1]. We selected the ChemBERTa model pre-trained on 10 M SMILES sequences for our analysis. Furthermore, we implemented and compared models using only a masked language modeling self-supervised learning approach. The pre-training step was completed within 10 epochs and the best-performing epoch was selected using the cross-entropy loss and AdamW optimizer. The maximum number of epochs was determined empirically by the preliminary experiments.

To conduct our analysis, we downloaded two versions of the original ChemBERTa that were uploaded to the HuggingFace hub: Seyonec and DeepChem. Subsequently, we fine-tuned our trained language models and the original ChemBERTa on the MoleculeNet benchmark datasets [31]. Considering the desirable drug properties, several medical chemistry application benchmark datasets were selected, including BACE, ESOL, Lipophilicity, BBBP, ClinTox, and Tox21. The specific description of each benchmark dataset is provided below.

The benchmark datasets used in this study for fine-tuning the language models were selected based on their relevance to desirable drug properties. These datasets include BACE, ESOL, Lipophilicity, BBBP, ClinTox, and Tox21. The BACE dataset contains binding results for a set of inhibitors of human β -secretase-1 (BACE-1), with quantitative or qualitative labels representing the half-maximal inhibitory concentration (IC₅₀). The ESOL dataset consists of 1,128 compounds with the water solubility measured in log-scaled mols in liters. The Lipophilicity dataset comprises 4,200 compounds with octanol/water distribution coefficients (log D at PH 7.4), a property that affects membrane permeability and solubility. The BBBP dataset contains 2,000 compounds with their prediction of blood-brain barrier penetration (BBBP). The ClinTox dataset includes 1,491

compounds with two statuses: clinical trial toxicity and FDA approval status. Finally, the Tox21 dataset contains 8,014 compounds with binary labels for 12 targets, including nuclear receptors and stress response pathways.

In this study, we utilized two multi-output datasets, ClinTox and Tox21, but selected only the CT-Tox and SR-p53 targets, respectively, to evaluate the performance of our models. To ensure model performance on a homogeneous dataset, we partitioned each dataset using an 8/1/1 ratio of training, validation, and test datasets, employing a scaffold splitter from the DeepChem library [32]. The data splitting ratio adopted following the values employed in the previous research [1]. A scaffold splitter was employed to split the dataset, considering the structural information of molecules. This approach enabled us to allocate an appropriate test set to measure the generalization performance of the models accurately. The dataset was split using the same random seed number.

For the fine-tuning step, we employed the AdamW optimizer and a cosine annealing learning rate scheduler with a learning rate of 1e-5 and a batch size of 32. Regression tasks were trained using a standard scaler, while classification tasks were trained using balanced sampling. Consistent with the original ChemBERTa paper, we used the root mean squared error (RMSE) as an evaluation metric for regression tasks and the area under the receiver operation curve (AUROC) score for classification tasks. We fine-tuned the models for 100 epochs.

C. MOLECULAR GENERATION

The generalized pre-trained transformer (GPT) architecture involves stacking multiple transformer decoder layers to generate sequences [33]. MolGPT is a variant of the GPT model specifically designed for molecular sequence generation [2]. By utilizing conditional flags, MolGPT can be trained as a conditional generator. Alternatively, when trained with-

out conditional flags, MolGPT operates as an unconditional generator. During pre-training, MolGPT employs causal language modeling (CLM), whereby the model generates the next token in a sequence based on the preceding tokens, one at a time [33]. Through CLM, the model can learn the conditional relationships between tokens, particularly in terms of their left-to-right order. Despite its relatively compact size (8 layers with 6 M parameters), MolGPT exhibits impressive performance in both unconditional and conditional molecular sequence generation tasks.

The original MolGPT was trained on two molecular generation benchmark datasets, namely MOSES and GuacaMol. The size of the vocabulary in each dataset was influenced by the number of unique characters employed in their respective representations, resulting in minor differences. As SMILES utilizes a more limited set of characters compared to natural language (more than 1 K), the vocabulary size for SMILES is relatively smaller (less than 100). Specifically, the MOSES dataset had a vocabulary size of 25, the GuacaMol dataset had a vocabulary size of 37, and the DM-Q and DM-QP datasets had a vocabulary size of 62. To perform the next-token prediction task, we trained four different versions of MolGPT for 10 epochs and the best-performing epoch was selected. The maximum number of epochs was determined empirically by the preliminary experiments and original experiments of MolGPT. The generated outputs of each trained MolGPT were sampled without any filtering, and the generation steps were conducted using unconditionally generation procedures. Each MolGPT generated 10 K sequences. Their performance was evaluated using several metrics: uniqueness, validity, logP, synthetic accessibility (SA) score, topological polar surface area (TPSA), QED score, Pfizer rule, and Lipinski rule. These metrics are defined as follows:

- Uniqueness: The degree to which the generated sequence is distinct from other generated sequences.
- Validity: The extent to which the generated sequence conforms to the SMILES format rules.
- LogP: The logarithmic partition coefficient, which measures the hydrophobicity of the molecule with respect to water and a nonpolar solvent. The optimal range for LogP is between 0 to 5 [34].
- Synthetic accessibility (SA) score: A measure of the difficulty of synthesizing a given compound. The SA score ranges from 1 (very easy to synthesize) to 10 (challenging to synthesize), and the optimal range is less than 6 [35].
- Topological polar surface area (TPSA): The sum of the surface area of all polar atoms. TPSA measures the drug permeability in cell membranes, and the optimal TPSA range is under $140 \text{ \AA}^2$ [36].
- QED score: A measurement of drug-likeness that ranges from 0 (not favorable) to 1 (favorable). The optimal range for the QED score is over 0.7 [37].
- Pfizer rule (3/75 rule): A rule that checks the likelihood of toxicity, where the optimal molecule satisfying the $\log P > 3$ and $\text{TPSA} < 75$ [27], [28].

- Lipinski rule: A rule that checks the likelihood of absorption and permeability, where the molecule could exhibit poor absorption if two properties are out of range. The properties include molecule weight ≤ 500 , $\log P \leq 5$, the number of hydrogen acceptors ≤ 10 , and the number of hydrogen donors ≤ 5 [38].

The inclusion of conditional tokens allows for managing the properties of generated molecules, including factors such as QED, logP, and TPSA. A conditional generation model was trained on conventional datasets utilizing QED, logP, and TPSA conditional tokens to assess the efficacy of an unconditioned generation model pre-trained on drug-like molecules. The MOSES and GuacaMol datasets were employed for training the conditional generation model, with the aforementioned conditions serving as filters for drug-like molecules. The conditional tokens were matched to the mean value of the unconditionally generated molecules to ensure consistency with the property density of unconditionally generated molecules. The Tanimoto similarity coefficient was utilized to evaluate the similarity between generated molecules, computed based on Equation 1. The Tanimoto similarity coefficient, also known as the Jaccard similarity coefficient, is defined as the ratio of the intersection of two sets to the union of the same two sets.

$$T(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (1)$$

III. RESULTS

A. MOLECULAR REPRESENTATION

The fine-tuning outcomes for the MoleculeNet benchmark dataset are presented in Table 1. Despite being trained on different pre-training datasets, the performance of ChemBERTa models was found to be comparable across various downstream tasks. Two baseline ChemBERTa models, namely Seyonec and DeepChem, are denoted as baselines, while the drug-like molecule-based pre-trained ChemBERTa models, DM-Q and DM-QP, are denoted as DMs. The performances of both the baselines and DMs were evaluated based on the average values obtained from each benchmark dataset. During the fine-tuning process using multiple benchmark datasets, the DMs exhibited a performance retention of over 90% compared to the baselines. However, it is noteworthy that the DMs demonstrated lower performance in regression tasks, specifically in the prediction of ESOL and Lipo, with benefits of 94.1% and 90.8% respectively. Intriguingly, the DMs exhibited significantly improved performance in predicting clinical toxicology, showing a benefit of 106.0% compared to the baselines.

B. MOLECULAR GENERATION

The present study investigated the property distribution of various molecular datasets, as illustrated in Figure 2. Notably, the GuacaMol dataset exhibited a more diverse distribution of properties compared to other datasets. Conversely, the drug-like molecules DM-Q and DM-QP presented a relatively narrow distribution, which can be attributed to the

TABLE 1. Fine-tuning performance of ChemBERTa pre-trained on drug-like molecules and the original ChemBERTa released by the DeepChem project. The classification tasks were evaluated in terms of AUROC, and the regression tasks were evaluated in terms of RMSE.

	Regression (RMSE) ↓			Classification (AUROC) ↑			
	BACE	ESOL	Lipo	BACE	BBBP	ClinTox	Tox21
Seyonec	1.0202	0.9804	0.6158	0.7222	0.7412	0.9370	0.7253
DeepChem	1.1856	1.0210	0.6102	0.7746	0.7206	0.9471	0.6950
DM-Q	1.1139	1.0485	0.6536	0.7801	0.7134	0.9986	0.6943
DM-QP	1.0212	1.0786	0.6966	0.7852	0.7278	0.9993	0.6772
Baselines	1.1029	1.0007	0.6130	0.7484	0.7309	0.9421	0.7102
DMs	1.0676	1.0636	0.6751	0.7827	0.7206	0.9990	0.6858
Benefit	103.3%	94.1%	90.8%	104.6%	98.6%	106.0%	96.6%

The arrows indicate the direction in which each metric is improving.

The baselines and DMs are average values of Seyonec and Deepchem, and DM-Q and DM-QP, respectively.

The benefits are the ratios of performance improvement between the values of baselines and DMs.

filtering process based on the QED score and Pfizer rule. The study also evaluated the performance of various MolGPT models in generating molecules based on different datasets, and Table 2 summarizes the results. As reported in the original study, the MolGPT models generated molecules with high validity and uniqueness. Moreover, MolGPT models trained on DM-Q and DM-QP datasets demonstrated superior performance in generating QED-acceptable molecules, achieving 94.25% and 94.0%, respectively.

In this study, we set the QED, logP, and TPSA values to 0.8, 3.95, and 42.4, respectively, for the conditional generation process. These values were matched to the properties of the unconditionally trained model on the DM-QP dataset. Overall, the conditional generation approach exhibited excellent performance in various metrics, with the exception of uniqueness. The Tanimoto similarity coefficients presented in Figure 3 demonstrated that the generated molecules using the conditional generation approach exhibited a relatively higher degree of structural similarity to each other. However, the uniqueness of the generated molecules was slightly lower than that of the unconditionally generated outputs. The mean value of the Tanimoto coefficient was 0.22, 0.25, and 0.25 for the unconditional generation, conditional generation trained on the MOSES dataset, and conditional generation trained on the GuacaMol dataset, respectively. This observation suggests that the model had difficulty in generating diverse molecules while adhering to conditional restrictions.

IV. DISCUSSION

Drug discovery is often described as finding a needle in a haystack. The process of identifying the desired target molecules from the vast molecular space is time-consuming and requires significant effort. To address these challenges, researchers are actively exploring in-silico methods, with a particular focus on the powerful sequence understanding capabilities offered by the transformers [1], [2], [3], [4], [5], [6], [7], [8]. Transformer models undergo a pre-training where they acquire knowledge about the characteristic features present in the sequences relevant to the domain of interest [1], [2], [4], [29], [33]. It has long been believed that pre-training with a large dataset encompassing a wide range of sequence features would result in improved performance

during subsequent fine-tuning. However, recent work by Krishna et al. has demonstrated that pre-training with a smaller dataset closely aligned with the target task can yield comparable or even superior performance [26].

Molecular sequences are often subjected to post-filtering processes aimed at selecting only those molecules possessing desirable properties. Alternatively, models can be trained by incorporating conditional terms. However, if the desirable molecular properties can be implicitly learned during the pre-training process, the need for post-filtering and conditional generation may be eliminated. In order to examine this hypothesis, we have defined a new dataset of molecules using the QED score and Pfizer rule as criteria. It is important to note that our study does not only focus on evaluating the superiority of the dataset itself but also assessing the performance of models trained using a pre-training dataset filtered based on desirable molecular properties for the target task.

In molecular representation, the performance of DMs maintained 90% of performances in each benchmark dataset compared to the baselines (Table 1). The DMs showed weakness in regression tasks (ESOL, Lipo), on the hand, showed strengths in classification tasks (BACE, ClinTox). Specifically, the benefit of DMs in ClinTox was 106.0%, however, the benefit of DMs in Tox21 was 96.6%. The Tox21 focuses on nuclear receptors and stress response pathways, the patterns were more heterogeneous compared to ClinTox. However, it is important to highlight that the DMs displayed a strong sensitivity toward clinical toxicology. In our experiment, we employed a straightforward approach by utilizing the Pfizer rule, a method that assesses the potential toxicity of a molecule. As a result, the DMs exhibited a heightened ability to accurately distinguish toxicology from a broad perspective. Moreover, there is potential for further enhancing the classification performance for specific target toxicology by incorporating more refined target screening measures.

In unconditional molecule generation, it has been observed that pre-training models using drug-like molecules produce more desirable outcomes (i.e., molecules meeting the QED and/or Pfizer rule criteria) compared to models pre-trained on MOSES and GuacaMol datasets. The MolGPT, due to its capabilities, maintains the inherent properties of

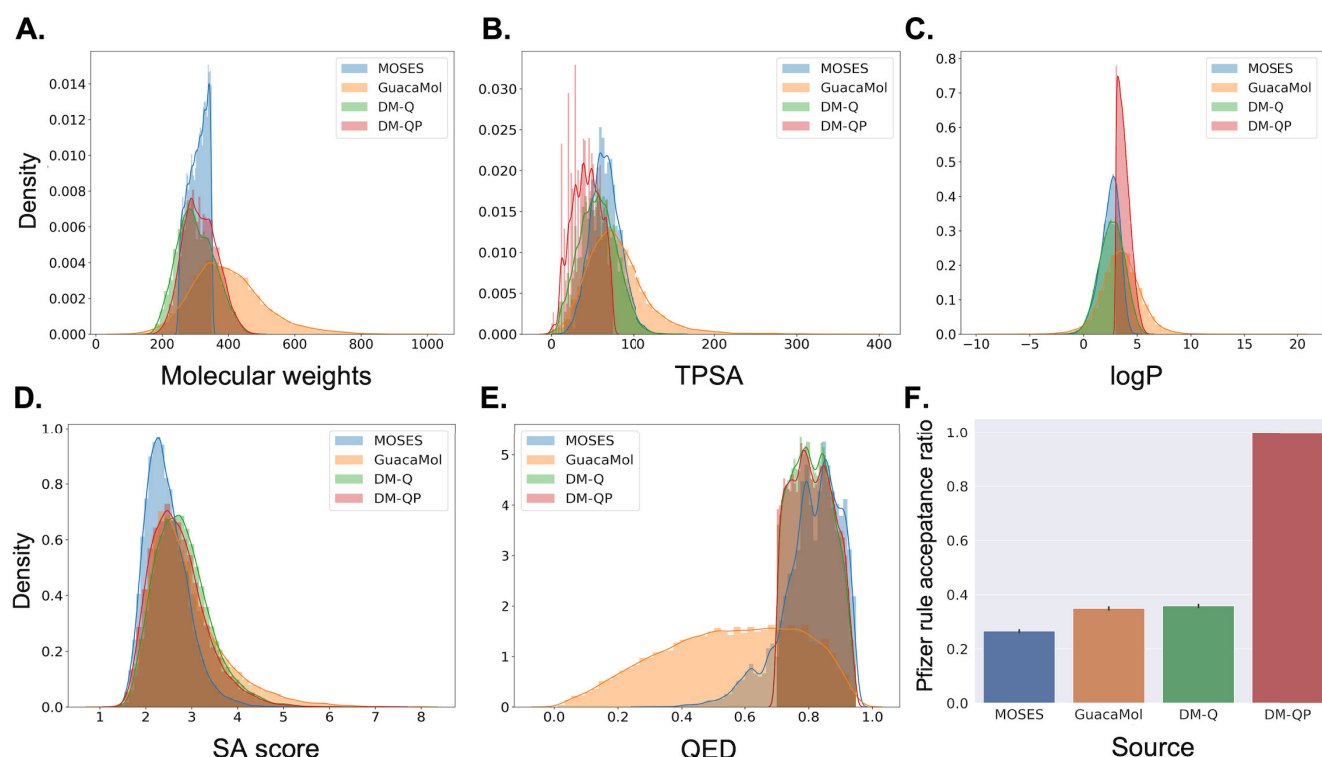


FIGURE 2. Probability distribution of molecular properties of pre-training datasets.

TABLE 2. Unconditional molecule generation performance of MolGPT. The proportion of the acceptance rate is summarized. The optimal range is described in the methods molecule generation section above.

	Pre-trained	Uniq	Validity	logP	SA	TPSA	QED	Pfizer	Lipinski
Unconditional	MOSES	99.79	99.75	99.98	100.00	99.96	88.84	27.51	100.00
	GuacaMol	99.62	99.14	80.35	99.74	92.57	30.34	36.97	89.78
	DM-Q	100.00	99.90	97.74	100.00	99.96	94.25	44.23	100.00
	DM-QP	99.94	99.97	94.15	99.99	99.91	94.02	94.80	100.00
Conditional	MOSES	85.70	99.22	100.00	100.00	100.00	98.40	99.98	100.00
	GuacaMol	97.20	99.21	99.94	99.93	100.00	91.94	99.94	100.00

Uniq, uniqueness; SA, synthetic accessibility score; TPSA, topological polar surface area; QED, the quantity of drug-likeness score; Pfizer, the Pfizer rule; Lipinski, the Lipinski rule;

the respective pre-training datasets in the outputs of each pre-trained model [2]. To further compare generation performance, we conducted conditional molecule generation using the MolGPT based on the MOSES and GuacaMol datasets. Similar to the unconditionally generated molecules, the conditional generation exhibited excellent performance in terms of QED and/or Pfizer rule acceptance rates, while displaying higher levels of duplication and structural similarity. This can be observed in Table 2 and Figure 3. The use of conditional tokens enables the control of the properties of generated molecules, but it simultaneously acts as a restriction that the model must consider. To address this, conditional GAN (cGAN) applies a diversity loss function to enforce the model to generate diverse outputs [39].

Considering the importance of diversity and ease of implementation, pre-training models on drug-like molecules could be a straightforward solution to implicitly teach the

model about desirable molecule properties. For instance, the multi-objective constrained molecule generation (MCMG) framework, which leverages reinforcement learning, filters final candidates with conditional flags, such as high QED and a low SA score [40]. In such scenarios, the utilization of conditional filtering can potentially introduce unintended consequences, such as reducing the diversity of the generated outcomes. Hence, pre-training models on drug-like molecules can serve as a means to transform conditional generation into unconditional generation. By incorporating drug-like molecule pre-training, the model becomes capable of generating molecules without the need for explicit conditions or constraints, thereby preserving a broader range of diversity in the generated results. Our approach to pre-training on drug-like molecules eliminates the need for conditional generation procedures and post-processing. Furthermore, the selection of the pre-training dataset can be filtered according to the

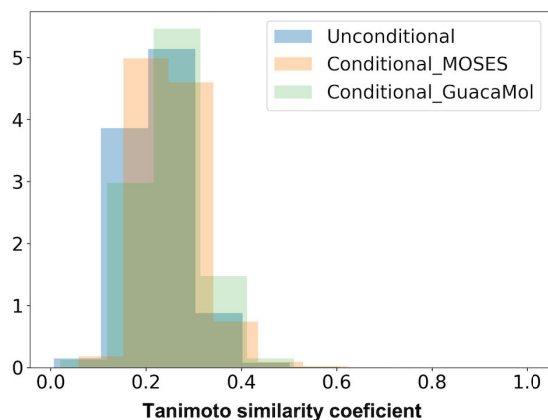


FIGURE 3. Distribution of the Tanimoto similarity coefficients.

researcher's objectives. Additionally, the drug-like molecule pre-training strategy is an agnostic approach to the model and training methodology, making it highly scalable.

V. CONCLUSION

This study demonstrated that drug-like molecules pre-training could yield considerable performances in molecular representations and generations. In terms of molecular representation, the ChemBERTa model trained on drug-like molecules achieved performance levels comparable to the baselines across various downstream tasks in the MoleculeNet benchmarks, encompassing a wide range of property datasets. Notably, the drug-like molecule-based models exhibited exceptional performance in predicting clinical toxicity.

Moving on to molecular generation, the models trained on drug-like molecules demonstrated enhanced desirability in terms of drug-likeness, structural diversity, and adherence to specific property ranges. When considering property matching, both the conditional generation approach and unconditional generation with drug-like molecule pre-training produced similar output properties. However, it is important to note that the conditional generation approaches resulted in a decrease in structural diversity. These findings underscore the potential of pre-training language models on targeted chemical spaces to augment their capacity for generating molecules possessing desirable properties. Through pre-training on filtered datasets, wherein the models are implicitly acquainted with desirable molecule properties, they can generate molecules that conform to specific criteria without necessitating complex conditional generation procedures.

In future studies, we aimed to find suitable assays of filters for the specific target and deeply investigate the trade-off between the diversity of pre-training datasets and more targeted scopes.

DATA AVAILABILITY

The drug-like filtered pre-trained datasets are uploaded on Zenodo data portal. The original ChemBERTa models are

1) Seyonec, and 2) DeepChem. The benchmark datasets that are utilized in molecular representation learning, can be found DeepChem. The source code of the original MolGPT can be found MolGPT GitHub.

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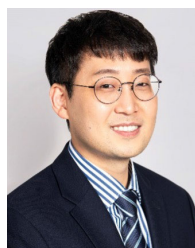
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